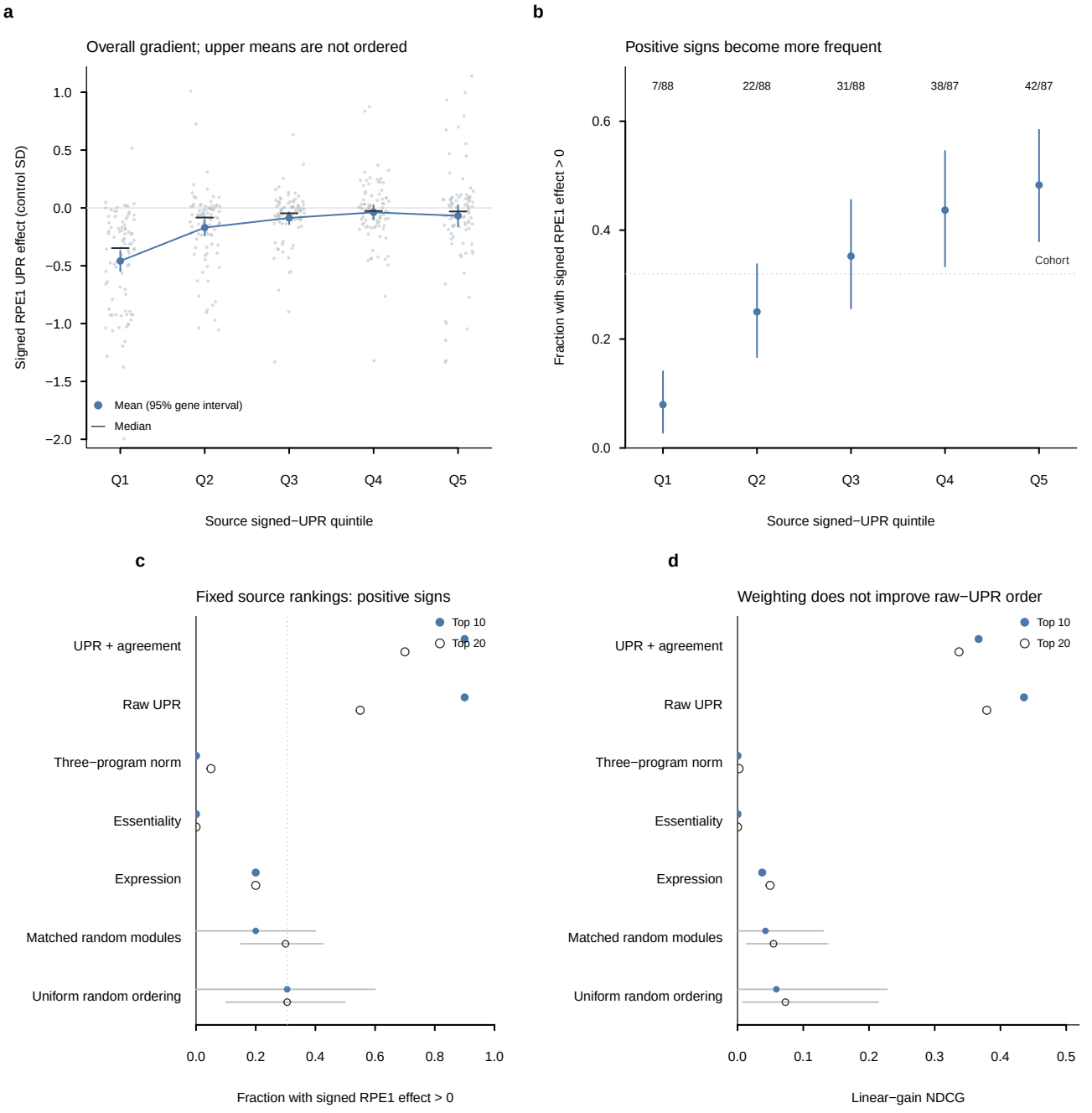
a
Overall gradient; upper means are not ordered
Signed RPE1 UPR effect (control SD)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Mean (95% gene interval)
Median
Q1
Q2
Q3
Q4
Q5
Source signed−UPR quintile
b
Positive signs become more frequent
Fraction with signed RPE1 effect > 0
7/88
22/88
31/88
38/87
42/87
0.6
0.4
0.2
0.0
Cohort
Q1
Q2
Q3
Q4
Q5
Source signed−UPR quintile
c
Fixed source rankings: positive signs
Top 10
Top 20
UPR + agreement
Raw UPR
Three−program norm
Essentiality
Expression
Matched random modules
Uniform random ordering
0.0
0.2
0.4
0.6
0.8
1.0
Fraction with signed RPE1 effect > 0
d
Weighting does not improve raw−UPR order
Top 10
Top 20
UPR + agreement
Raw UPR
Three−program norm
Essentiality
Expression
Matched random modules
Uniform random ordering
0.0
0.1
0.2
0.3
0.4
0.5
Linear−gain NDCG